A Formal Physics Laboratory Report

REPORT LAB 08

MEASURING REFRACTIVE INDEX OF GLASS USING MICROSCOPE

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I. Aims/Purposes:

- Determining refractive index of the glass by microscope

II. Method, Equipment and Procedure:

1) *Equipment*

- Microscope which has objective lens' magnification at x4, x10, x100 and eyepieces lens magnification at x10, x16
- Panme ruler 0 ÷ 25mm, precision 0.01mm.
- Thin film that needs to measure refractive index.

2) *Method*

- Indirect measurement

3) *Procedure*

- Using the Panme ruler to determine the true thickness.
- Using a microscope, determine the effective thickness of a thin coating.

- Step 1: Measuring the real thickness of thin film by using Panme ruler
- Step 2: Moving the stage down to the lowest position
- Step 3: Rotate the dial until you witness the upper surface and a sharp image of the film. Writing down the position of the ring ruler corresponding to the mark L_0 on the opposite site of the mark microscope body
- Step 4: Adjusting the small dial until you see the low surface's image. Remember to count the number of rounds N of the ring ruler whenever passing the mark L_0. Read and write the final position of the ring ruler

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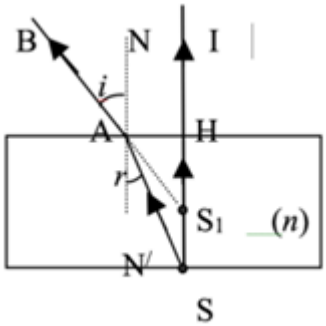
III. Equations



$$tgAS_1H = \frac{AH}{S_1H} = tgNAB = tgi \quad (1)$$

$$tgASH = \frac{AH}{SH} = tgN'AS = tgr \quad (2)$$

$$sini = tani\,,\ sini \approx tanr \qquad (3)$$

Dividing (1) by (2), and taking into account (3), we have:

$$\frac{d}{d_1} = \frac{SH}{S_1H} \approx \frac{sini}{sinr} = n$$

$$=> \bar{n} = \frac{\bar{d}}{\bar{d}_1}$$

$$\Delta d = \Delta d_{sys} + \Delta\bar{d}$$

$$\Delta d_1 = \Delta d_{1_{sys}} + \Delta\bar{d}_1$$

$$\frac{\Delta n}{\bar{n}} = \frac{\Delta d}{\bar{d}} + \frac{\Delta d_1}{\bar{d}_1}$$

$$d_1 = 0.2N + 0.001\left(l - l_0\right) mm \qquad if\ l > l_0$$

$$d_1 = 0.2N + 0.001\left(l + 200 - l_0\right) mm \qquad if\ l < l_0$$

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IV. Experimental data

Table 1

- Accuracy of the micrometer caliper: 0.01 (*mm*) - Accuracy of the microscope's rotary ruler: 0.002(*mm*)									
	Effective thickness d_1 (*mm*)					Real thickness d (*mm*)			
	N	l_0	l	d_1	Δd_1	k	m	d	Δd
1	8	14	90	1.676	0.0252	5	36	2.86	0.002
2	8	152	64	1.712	0.0108	5	37	2.87	0.008
3	8	128	32	1.704	0.0028	5	36	2.86	0.002
4	8	30	178	1.748	0.0468	5	36	2.86	0.002
5	8	150	16	1.666	0.0352	5	36	2.86	0.002
Avg.				1.7012	0.0242			2.862	0.003

V. Calculation works

Refractive index:

$$\overline{n} = \frac{\overline{d}}{\overline{d_1}} = \frac{2.862}{1.7012} = 1.6823$$

Uncertainties:

$$d = d_{sys} + \overline{d} = 0.01 + 0.003 = 0.013\ (mm)$$

$$d_1 = d_{1sys} + \overline{d_1} = 0.002 + 0.0242 = 0.0262\ (mm)$$

$$\frac{\Delta n}{\overline{n}} = \frac{\Delta d}{\overline{d}} + \frac{\Delta d_1}{\overline{d_1}} = \frac{0.013}{2.862} + \frac{0.0262}{1.7012} = 0.0199$$

$$=> \Delta n = 0.0199\ x\ 1.6823 = 0.0335$$

Result:

$$n = \overline{n} \pm \Delta n = 1.6823 \pm 0.0335$$

VI. Questions

1. *State the law of refraction. Draw the illustration image.*

- The law of refraction:
  + The incident ray, reflected ray and the normal, to the interface of any two given mediums, all lie in the same plane.

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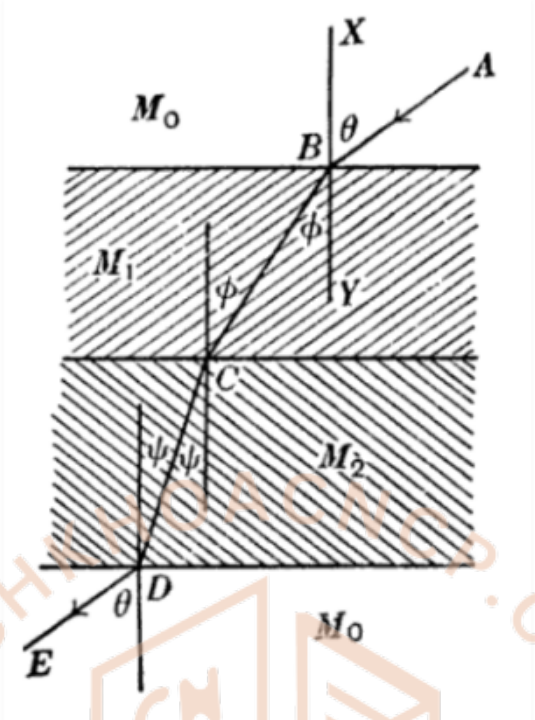
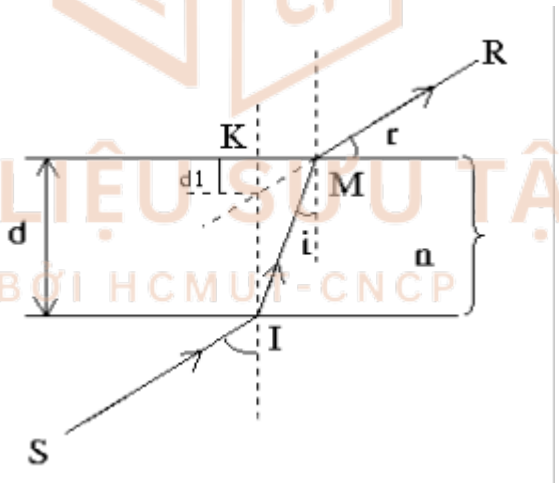
+ The ratio of the sine of the angle of incidence and sine of the angle of refraction is constant. This is also known as Snell's law of refraction



2. *Explain the mechanism of image forming of a light point when viewing through the parallel thin film. Show the properties of the image*



- The properties of the image:
  + A ray of light parallel to a ray of light
  + Real objects give virtual images and virtual objects give real images
  + Object and image are parallel and have the same magnitude

3) Define the effective thickness of the thin film. Show the relation of refractive index of the given thin film and its effective.

- In internal reflection spectroscopy effective thickness is defined as the thickness of material required to give a spectrum of the same contrast in a transmission measurement as that obtained via internal reflection.

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- The relation of refractive index of the given thin film and its effective: (according to the picture in question number 2)

$$n = \frac{KI}{KI_1} = \frac{d}{d_1}$$

4) Describe the components of the panme ruler and how to use it.

- Panme ruler includes:
  + Lock
  + Sleeve
  + Thimble
  + Ratchet knob
  + Anvil
  + Spindle
  + Frame
- How to use it:
  + Step 1: When measuring with the left hand, hold the micrometer, with the right hand, turn the measuring head to come in contact, then turn the knob to make the measuring head contact the object with the correct measuring pressure.
  + Step 2: Must keep the centerline of the two measuring torches coincide with the size of the object to be measured
  + Step 3: In case the micrometer must be removed from the measuring position, turn the lock nut (brake lever) to fix the dynamic measuring head before removing the micrometer from the measuring object.
  + Step 4: When measuring against the edge of the moving ruler, we can read the "mm" and half "mm" of the size on the main ruler.
  + Step 5: Based on the standard line on the main ruler, we can read the percentage "mm" on the secondary ruler (the value of each line is 0.01 mm)

- In the process of using a micrometer or any other measuring tool, we need to pay attention to how the ruler is stored to keep the measurement accurate.
  + Do not use the micrometer to measure while the object is rotating.
  + Do not measure rough, dirty surfaces.
  + The measuring object must be cleaned before measuring.
  + Do not screw the auxiliary ruler screw directly so that the torch is pressed against the measuring object.
  + It is necessary to limit taking the Micrometer out of the measuring position to read the size
  + The measuring faces of the Micrometer need to be handled with care to avoid rust and abrasive sand, grit or metal workpieces.

5) Describe the microscope and how to use it to measure the effective thickness of the thin film

- The microscope includes:
  + Ocular
  + Objective adjustment price
  + Objective lens

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    + Lighting systems
    + Specimen order price
    + Converging lenses and aperture system
    + Fine tuning
- How to use it:
    + Step 1: Place the specimen on the table for the specimen, then use clamps to fix the specimen. Depending on the observation sample as well as the purpose of the sample, choose the appropriate objective lens. With the 100x objective, add 1 drop of lubricating oil to submerge on the slide.
    + Step 2: Adjust the condenser, specifically as follows: With the 10x objective, lower the condenser to the end, and with the 40x objective, leave it in the middle. Then adjust the screen size to fit the objective.
    + Step 3: Lower the objective lens close to the eye to see the specimen. The eye looks into the eyepiece and the hand turns the macro screw to bring the object up until a blurry image of the microfield is visible, then adjusts the micro level screw to make the image clear.
    + Step 4: Repair, observe, ... the specimen and make notes if necessary.
    + Step 5: After using the optical microscope, disconnect the power and clean the glass.

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